

vii. Origin of the Earth

*Earth is one of the planets in the solar system. Its origin is not different from that of the other planets. Different scholars have propounded different theories regarding the subject. *One of the earliest works on the subject was presented by **Immanuel Kant**. He propounded the theory of nebulous material. * In 1755 A.D. Immanuel Kant presented the Gaseous hypothesis. Later in 1796 A.D., **Laplace** presented his research on the topic which came to be known as **Nebular Hypothesis**. In 1905 A.D. **Chamberlin** and **Moulton** proposed the **Chamberlin-Moulton planetesimal hypothesis** to describe the formation of the solar system. The theory was based on the idea that a star passed close enough to the sun early in its life to cause tidal bulges to form on its surface, causing the material to be ejected repeatedly from the sun. Due to the gravitational effects of the passing star, materials would cool and condense into numerous small bodies, which they called planets. *In 1919 the famous British scientist **James Jeans** propounded the **Tidal Hypothesis** and another scientist **Sir Harold Jeffery** made some corrections to it. After some years, the theory of a wandering star around the sun started getting acceptance. The theory was called **Binary Theories**. *In 1943, Otto Schmidt propounded the interstellar dust hypothesis. He proposed that the sun, in its present form, passed through a dense interstellar cloud and emerged enveloped in a cloud of dust and gas. **Otto Schmidt** of Russia and **Carl Weizsacker** of Germany revised the Nebular Hypothesis.

They believed dust like 'Solar Nebula' with heavy helium and hydrogen surrounds the sun. A disk-shaped cloud was created by the friction of particles in the solar Nebula. The planet was formed due to the accretion process.

*In modern times the most accepted theory regarding the origin of Universe is the **Big Bang Theory** also known as **expanding Universe Theory**. Universal expansion was proved by **Sir Edwin Hubble** in 1920. According to the Big Bang Theory, a violent explosion of a tiny body occurred which resulted in huge expansion and the change of some particles into an energy form. Due to this expansion, some energy got converted into matter. This happened 13.7/13.8 billion years ago. *Another proof of this Edwin Hubble concluded that galaxies are drifting **apart**. This is important observational evidence consistent with the hypothesis of an expanding Universe. After initial expansion, the universe cooled sufficiently to allow the formation of subatomic particles and later simple atoms. Giant clouds of these elements later combined due to gravity eventually forming early stars and galaxies. The galaxies further broke down resulting in the formation of stars and stars broke down (ran out of fuel) resulting in the formation of early planets.

*In 1907 John Joly, used his knowledge of mineralogy to figure out the age of different minerals. According to him certain rock-forming minerals were often characterized by the presence of small circular dark spots or concentric rings. He called it the **Mineral Halo**. *Professor **H.H. Russell** used Uranium –Lead and Thorium-Lead to determine the age of Earth. According to it the age of our planet is 2 billion and 25 crore years and 4 billion and 60 crore year old respectively. *Radiometric dating has proven to be the most successful way of figuring out the Earth's age. As Uranium and Thorium are found in one form or the other in different quantities in every rock formation, the Uranium Dating Technique to determine the age of Earth becomes very useful.

1. Which of the following scholar has suggested that the Earth has originated from the gases and the dust particles ?

(a) James Jeans (b) H. Alfven
(c) F. Hoyle (d) O. Schmidt

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Ans. (d)

James Jeans and Harold Jeffreys are related to the tidal hypothesis of the origin of Earth. F. Hoyle and Littleton presented the Nova hypothesis and Otto Schmidt gave the hypothesis of gas and dust for the genesis of Earth (Interstellar dust hypothesis). According to this hypothesis, large quantities of gas and dust particles captured by the gravitational force of the sun formed the Earth.

2. Which of the following Method is used to determine the age of the Earth ?

(a) Carbon dating for the age of the fossils.
(b) Germanium dating
(c) Uranium dating
(d) All of the above

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Ans. (c)

The uranium dating method is used for determining the age of Earth.

viii. The Geological History

*Geological History of Earth follows the major events in Earth's past based on the geological time scale. The first efforts to understand the Geological History of Earth were made by a French naturalist Comte de Buffon. At present the geological time scale is used, which is a system of chronological dating that relates geological states to time.

*The primary defined divisions of time are eons. Eons are divided into eras which are in turn divided into periods and epochs.